

BOL'SHAYA MURTA, Russian Federation

WMO: 294710

Lat: 56.8950N

Lon: 93.1650E

Elev: 180

StdP: 99.18

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.4	-35.8	-41.8	0.1	-39.1	-38.4	0.1	-35.7	6.5	-9.3	6.0	-7.8	0.1	310	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.1	18.8	27.4	18.2	25.6	17.4	20.6	25.9	19.6	24.8	18.6	23.7	2.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.0	22.6	17.7	13.0	21.8	16.7	12.1	21.0	60.4	25.9	56.9	24.6	53.6	23.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.2	4.5	-40.8	32.2	4.3	1.2	-43.9	33.1	-46.4	33.8	-48.9	34.5	-52.0	35.4		
(5)			-40.1	22.2	4.1	1.3	-43.1	23.1	-45.5	23.9	-47.9	24.6	-50.9	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.5	-19.4	-15.3	-6.9	2.1	9.3	16.7	18.8	15.8	8.6	0.9	-9.6	-16.4		
	DBStd	14.68	9.84	8.38	6.89	5.14	5.17	4.15	3.07	3.69	4.38	5.44	8.60	9.48		
	HDD10.0	4241	911	709	525	242	78	3	0	3	78	284	589	818		
	HDD18.3	6620	1169	942	783	488	282	78	32	95	293	542	839	1076		
	CDD10.0	757	0	0	0	3	57	204	273	182	36	1	0	0		
	CDD18.3	94	0	0	0	0	3	29	46	16	1	0	0	0		
	CDH23.3	948	0	0	0	2	47	346	382	159	13	0	0	0		
	CDH26.7	219	0	0	0	0	8	91	84	34	2	0	0	0		
	WSAvg	1.8	1.4	1.6	2.0	2.3	2.4	1.7	1.5	1.5	1.7	1.9	1.9	1.6		
	PrecAvg	465	20	17	17	24	39	61	70	66	49	36	37	29		
(15) Precipitation	PrecMax	580	40	38	43	44	93	191	154	119	88	55	60	54		
	PrecMin	325	3	6	3	6	10	8	3	26	12	16	18	5		
	PrecStd	68	8	7	10	11	21	36	38	25	20	10	11	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	2.3	9.2	19.8	27.9	31.6	31.1	29.9	25.3	16.4	7.7	1.9		
		MCWB	-1.6	-0.4	3.8	10.2	15.7	18.4	20.1	19.5	16.1	9.6	5.0	-0.3		
	2%	DB	-1.6	0.2	6.3	16.1	24.5	29.1	28.9	27.3	20.9	12.5	5.0	-0.3		
		MCWB	-2.8	-1.8	2.1	8.6	14.1	18.0	19.6	18.9	14.5	8.1	2.8	-1.6		
	5%	DB	-3.4	-1.4	4.3	13.2	21.8	27.2	27.2	24.9	18.3	10.3	3.0	-2.1		
		MCWB	-4.6	-2.9	1.0	6.9	12.8	17.4	19.0	17.7	13.0	7.3	1.2	-3.2		
	10%	DB	-5.6	-3.4	2.4	10.6	19.1	25.0	25.5	22.9	15.7	8.2	1.1	-4.1		
		MCWB	-6.6	-4.8	-0.3	5.3	11.5	16.5	18.2	16.8	11.8	5.8	-0.1	-5.1		
	0.4%	WB	-1.2	-0.1	4.2	11.0	16.9	20.7	22.3	21.4	16.9	10.7	5.7	0.1		
		MCDB	0.1	1.5	8.0	17.1	25.5	26.6	27.6	26.5	22.7	14.1	7.2	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.7	-1.5	2.6	9.1	15.1	19.5	21.0	19.8	15.1	9.0	3.0	-1.5		
		MCDB	-1.6	0.0	5.4	15.1	22.0	25.9	26.3	24.9	19.9	11.5	4.7	-0.4		
	5%	WB	-4.5	-3.0	1.2	7.3	13.6	18.5	20.1	18.6	13.8	7.6	1.3	-3.2		
		MCDB	-3.5	-1.4	3.9	12.5	20.4	25.0	25.1	23.4	17.2	10.1	2.8	-2.0		
	10%	WB	-6.5	-4.8	-0.1	5.6	12.1	17.6	19.1	17.5	12.4	5.8	-0.1	-5.1		
		MCDB	-5.5	-3.4	2.3	10.1	18.3	23.6	23.8	21.9	15.2	8.1	1.0	-4.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	9.3	10.5	9.9	11.1	11.0	10.1	10.0	9.0	7.1	6.8	7.5		
		MCDBR	6.4	8.4	11.0	15.1	15.8	15.2	13.0	13.6	14.0	10.9	6.5	8.3		
		MCWBR	5.8	7.1	7.8	8.5	7.7	5.7	5.3	5.9	7.6	6.8	5.3	7.6		
		MCDBR	6.2	7.9	9.8	13.9	13.7	12.2	10.3	11.5	12.3	10.4	6.2	7.9		
	5% WB	MCWBR	5.7	7.0	7.3	8.3	7.3	5.3	4.9	5.7	7.2	6.8	5.3	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.318	0.360	0.390	0.410	0.415	0.436	0.411	0.367	0.348	0.315	0.281		
	taud		1.998	1.985	1.951	2.117	2.206	2.279	2.230	2.273	2.354	2.238	2.044	1.949		
	Ebn at Noon		545	693	765	816	826	827	796	791	776	669	500	424		
	Edh at Noon		64	99	134	133	130	123	127	114	91	77	61	51		
(44) All-Sky Solar Radiation	RadAvg		0.71	1.69	3.05	4.37	4.95	5.98	5.42	4.26	2.64	1.34	0.65	0.41		
	RadStd		0.09	0.12	0.21	0.29	0.51	0.52	0.36	0.35	0.35	0.11	0.10	0.07		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page